

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461038.912 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.044
Transit depth (Rp/Rs)^2	2.3 +/- 1.33 [%]
Orbital Inclination (inc)	87.4 +/- 2.49
Airmass coefficient 1 (a1)	1.123 +/- 0.03
Airmass coefficient 2 (a2)	-0.069 +/- 0.041
Scatter in the residuals of the lightcurve fit is	2.66 %
Best Comparison Star	#3 - [185, 269]
Optimal Aperture	4.27
Optimal Annulus	9.25
Transit Duration (day)	0.104 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.152 $\pm$ 0.044 vs 0.11 (deviation 0.75 σ)
Duration fit vs published	0.104 d vs 0.1208 d (deviation 0.49 σ)
Mid-time O-C	-25.9 min
Depth SNR	2.7
Residual scatter	2.66 %

Light curve

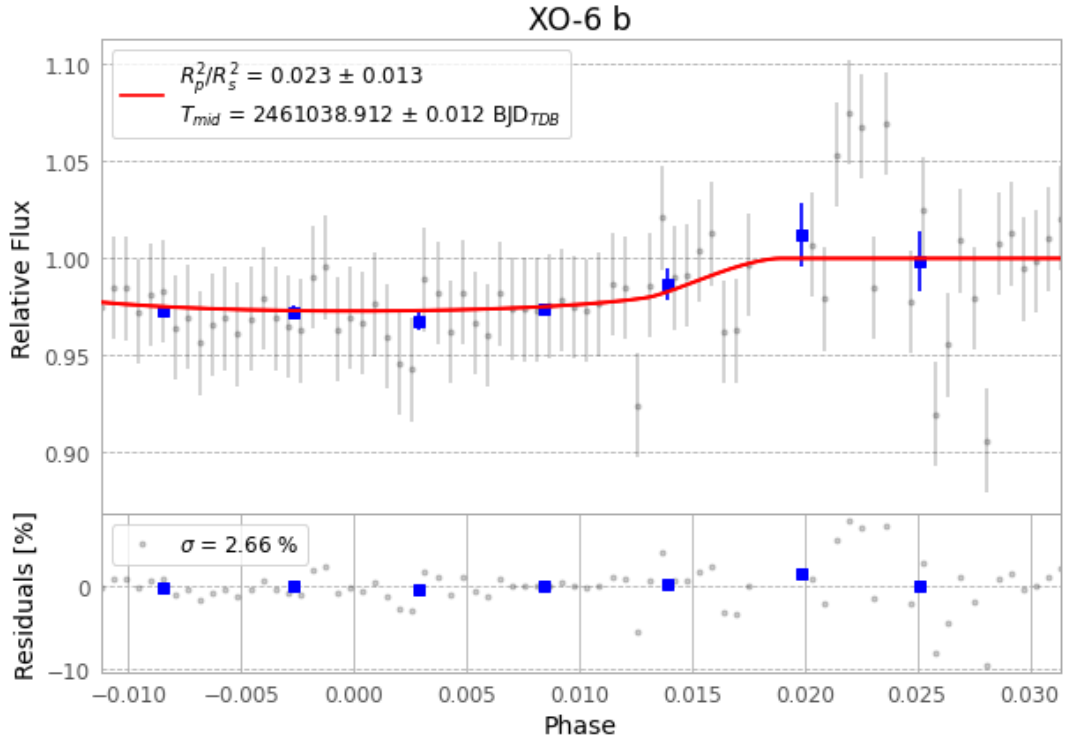


Figure 1 — FinalLightCurve_XO-6 b_2025-12-29.png (SHA-256 088f893600e369f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 360], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52490c045b8cd867...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

78 FITS frames (51 MB), XO-6251229084516.FITS → XO-6251229123615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-251229.log (fc6940e1cc94...), FinalLightCurve_XO-6 b_2025-12-29.png (088f893600e3...), FinalLightCurve_XO-6 b_2025-12-29.csv (c643f2c89a56...)

Compute resources

Wall time 182.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:18Z.